

NLP User Research Project

Automated review intelligence using
scraping, NLP classification, and live
dashboards

Trustpilot · Apify

Python · NLP

Dash · Excel

CONTEXT



The Situation

Critical part of Product Roles is User Research



It has to be based on data as much as possible — Because it
effects all following actions



This Project is about making the analyzing automated
& clear using NLP

ARCHITECTURE



The Pipeline

Input — Company Product & Review Site



Scraping
Reviews Text

automation-lab actor

Scraping
Stars Review

blackfalcondata actor

Merge &
Deduplicate
dedup by exact text

Classification
Good & Bad Experience
stars → sentiment

Tag Each
Review
by Category
first match wins

Saves
Dataset
balanced .xlsx



Output — Excel & Dashboard

⚡ 2 Apify actors

🔄 7 pipeline steps

📊 Balanced 50/50 output



Pipeline Explained

1 Scraping Reviews Text

automation-lab~trustpilot

- Fetches **mixed-star reviews**, sorted by recency
- Filter: **English language only** — removes non-EN noise
- Memory auto-scaled: **1 GB ≤ 1k reviews**, up to 4 GB
- Timeout: 10 minutes per run

2 Scraping Stars Review

blackfalcondata actor

- Targeted pull of **1-star reviews only**
- Ensures sufficient **negative data coverage**
- Stars passed as integers (actor-specific requirement)
- Timeout: up to **3 hours** for high-volume pulls



3 Merge & Deduplicate

Combines both scrape batches into one clean list

- Merges general reviews + 1-star reviews into a single dataset
- Removes duplicates by **exact text match** — no review counted twice
- Extracts text from fields: **text / body / reviewBody / content / review**
- Drops items under 5 characters (noise removal)



4 Classification — Good and Bad Experience

Star rating first · keyword sentiment fallback

- 4–5 stars** → Good Experience | **1–2 stars** → Bad Experience
- 3 stars or no rating** → keyword fallback:
 - Negative signals ("scam", "terrible", "worst", "never again") → Bad
 - No negative signal → Good (default assumption)



5 Tag Each Review by Category

First keyword match wins · Bad tags checked in priority order

✓ Good Tags

- Fast Delivery
- Responsive Support
- Ease of Use
- Product Effectiveness
- General Positive (fallback)

✗ Bad Tags (priority order)

- Takes Too Long
- Website Issue
- Payment Issue
- Customer Service
- Product Quality (fallback)



6 Saves Dataset — Balanced Excel (.xlsx)

Equal Good/Bad split · Styled rows · Frozen header

- Trims larger class to match smaller: **Good count = Bad count** (50/50 balance)
- Columns: **DOCTOR/PATIENT · ORIGINAL REVIEW · CLASSIFICATION · TAG**
- Good rows → light green fill | Bad rows → light red fill | Header → dark navy

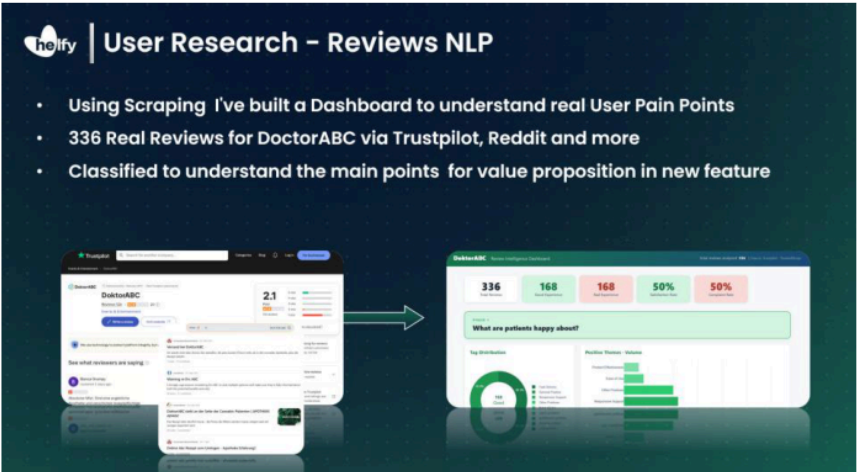


7 Output — Excel & Live Dashboard

Plotly Dash · Auto-opens browser · localhost:8050

- Global KPI bar:** Total Reviews · Good · Bad · Satisfaction % · Complaint %
- Phase 1 — Good (green):** donut chart + horizontal bar chart + sample positive review cards
- Phase 2 — Bad (red/orange):** donut chart + horizontal bar chart + sample complaint cards

Real Example:



The data after scraping:

TAG	CLASSIFICATION	ORIGINAL REVIEW	DOCTOR/PATIENT
General Positive	Good Experience	Alles hat bestens geklappt! Vielen Dank. Top (Anonymous — Trustpilot, 2026-08-15)	Patient
Fast Delivery	Good Experience	It's really good the delivery is fast and also the medication is really bio and healthy (Anonymous — Trustpilot, 2026-08-15)	Patient
General Positive	Good Experience	Alles super! (Anonymous — Trustpilot, 2026-08-14)	Patient
General Positive	Good Experience	Out of all the websites to order cannabis from, doktorabc.com is the most hassle free one. From getting the prescription to approval usually less than 90 minutes and we just pick up at our local pharmacy. We Love the selection and being able to pick it up ourselves. Gonna miss this when we get back to the US. (Anonymous — Trustpilot, 2026-08-14)	Patient
Ease of Use	Good Experience	Easy to operate the app and to communicate, came in 2 days, perfect experience (Anonymous — Trustpilot, 2026-08-14)	Patient
Fast Delivery	Good Experience	Good product and fast delivery! (Anonymous — Trustpilot, 2026-08-14)	Patient
Fast Delivery	Good Experience	Fast process, clear application	Patient
Website Issue	Bad Experience	Lots of communication about what is happening in the process Quick and positive result. (Anonymous — Trustpilot, 2026-08-14)	Patient
Fast Delivery	Good Experience	There are some strains I like, however, getting to find them is not forthcoming. Information and effects sometimes differ way to much to be a reliable source. One has to test him or herself and more often than not, very disappointing. (Anonymous — Trustpilot, 2026-08-14)	Patient
Responsive Support	Good Experience	Order arrived home in less than 1 hour. The Uber drivers were super professional and polite. Super positive experience :) (Anonymous — Trustpilot, 2026-08-13)	Patient
Product Effectiveness	Good Experience	good service and regular updates (Anonymous — Trustpilot, 2026-08-13)	Patient
Fast Delivery	Good Experience	100% trustable, great UX/UI, high quality vs. good price (Anonymous — Trustpilot, 2026-08-13)	Patient
Fast Delivery	Good Experience	Very quick services, multiple varieties and good quality products. (Anonymous — Trustpilot, 2026-08-13)	Patient
General Positive	Good Experience	Best dealer in der Stadt (Anonymous — Trustpilot, 2026-08-13)	Patient
General Positive	Good Experience	perfect und qualitative sehr empfehls wert, (Anonymous — Trustpilot, 2026-08-13)	Patient

Result :

